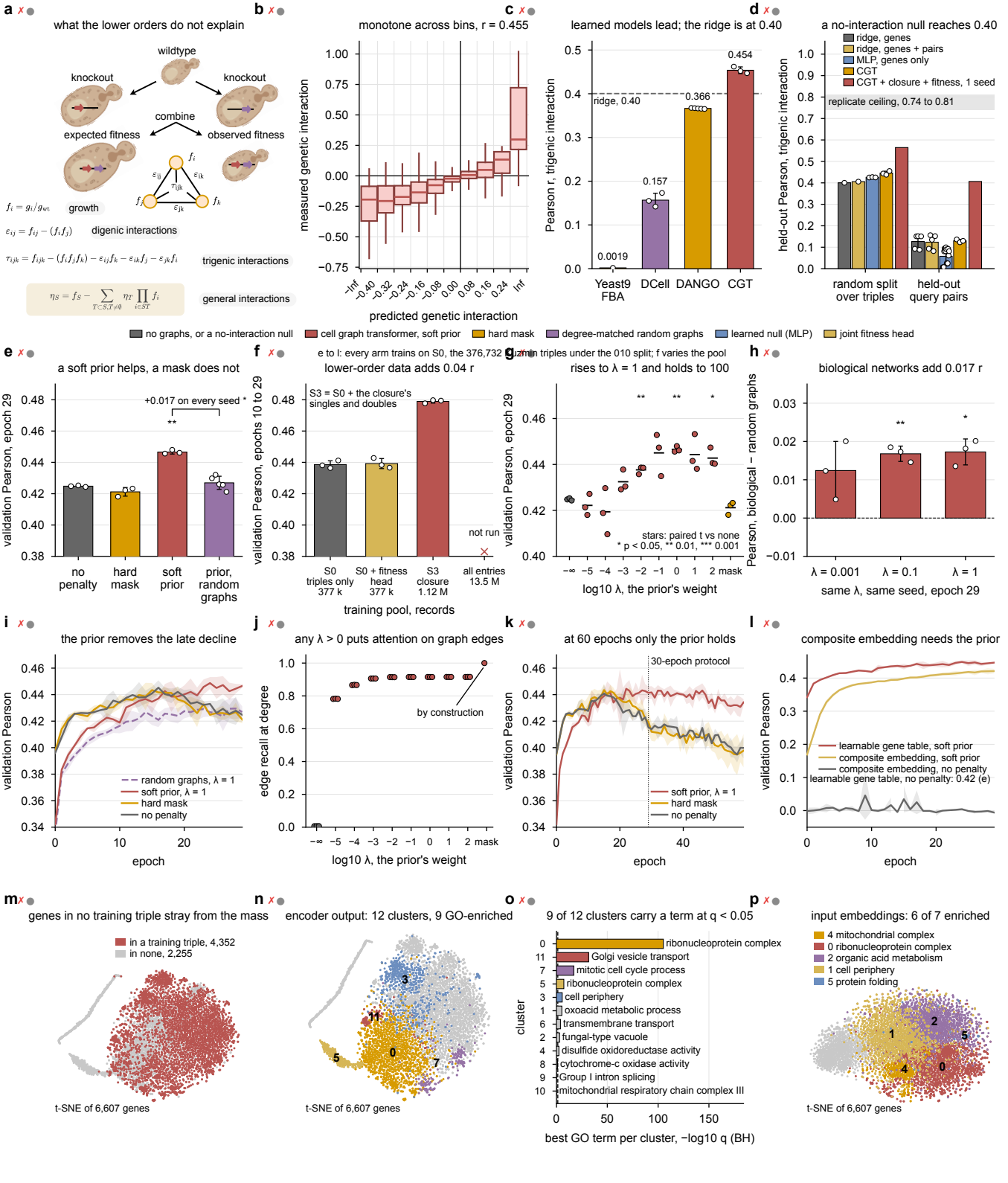

a
what the lower orders do not explain
wildtype
knockout
knockout
combine
expected fitness
observed fitness
growth
digenic interactions
trigenic interactions
general interactions
b
monotone across bins, r = 0.455
measured genetic interaction
predicted genetic interaction
c
learned models lead; the ridge is at 0.40
Pearson r, trigenic interaction
ridge, 0.40
0.0019
0.157
0.366
0.454
Yeast9 FBA
DCell
DANGO
CGT
d
a no-interaction null reaches 0.40
held-out Pearson, trigenic interaction
ridge, genes
ridge, genes + pairs
MLP, genes only
CGT
CGT + closure + fitness, 1 seed
replicate ceiling, 0.74 to 0.81
random split over triples
held-out query pairs
no graphs, or a no-interaction null
cell graph transformer, soft prior
hard mask
degree-matched random graphs
learned null (MLP)
joint fitness head
e
a soft prior helps, a mask does not
validation Pearson, epoch 29
+0.017 on every seed *
**
no penalty
hard mask
soft prior
prior, random graphs
f
lower-order data adds 0.04 r
validation Pearson, epochs 10 to 29
S3 = S0 + the closure's singles and doubles
not run
S0 triples only 377 k
S0 + fitness head 377 k
S3 closure 1.12 M
all entries 13.5 M
training pool, records
g
rises to λ = 1 and holds to 100
validation Pearson, epoch 29
stars: paired t vs none
* p < 0.05, ** 0.01, *** 0.001
log10 λ, the prior's weight
h
biological networks add 0.017 r
Pearson, biological − random graphs
λ = 0.001
λ = 0.1
λ = 1
same λ, same seed, epoch 29
i
the prior removes the late decline
validation Pearson
random graphs, λ = 1
soft prior, λ = 1
hard mask
no penalty
epoch
j
any λ > 0 puts attention on graph edges
edge recall at degree
by construction
log10 λ, the prior's weight
k
at 60 epochs only the prior holds
30-epoch protocol
validation Pearson
soft prior, λ = 1
hard mask
no penalty
epoch
l
composite embedding needs the prior
validation Pearson
learnable gene table, soft prior
composite embedding, soft prior
composite embedding, no penalty
learnable gene table, no penalty: 0.42 (e)
epoch
m
genes in no training triple stray from the mass
in a training triple, 4,352
in none, 2,255
t-SNE of 6,607 genes
n
encoder output: 12 clusters, 9 GO-enriched
t-SNE of 6,607 genes
o
9 of 12 clusters carry a term at q < 0.05
cluster
ribonucleoprotein complex
Golgi vesicle transport
mitotic cell cycle process
ribonucleoprotein complex
cell periphery
oxoacid metabolic process
transmembrane transport
fungal-type vacuole
disulfide oxidoreductase activity
cytochrome-c oxidase activity
Group I intron splicing
mitochondrial respiratory chain complex III
best GO term per cluster, −log10 q (BH)
p
input embeddings: 6 of 7 enriched
4 mitochondrial complex
0 ribonucleoprotein complex
2 organic acid metabolism
1 cell periphery
5 protein folding
t-SNE of 6,607 genes